

Dataset for ROP Decision System

- **Zone: Zone 1, Plus: Plus, Stage: Stage 3, Decision: Treatment** In Retinopathy of Prematurity (ROP), Zone 1 involvement with Plus disease and Stage 3 indicates aggressive posterior ROP, characterized by extraretinal fibrovascular proliferation (EFP) along with vascular dilation and tortuosity. This meets Type 1 ROP criteria per ETROP guidelines, posing a high risk of rapid progression to retinal detachment and potential blindness if untreated. Decision: Treatment within less than 1 week, typically involving peripheral retinal ablation via laser photocoagulation or intravitreal anti-VEGF injections such as bevacizumab; urgent referral to a pediatric ophthalmologist is essential, with post-treatment monitoring every 5-7 days to assess regression and complications like macular dragging.
- **Zone: Zone 1, Plus: Plus, Stage: Stage 2, Decision: Treatment** ROP in Zone 1 at Stage 2 with Plus disease features a ridge of elevated tissue at the vascular-avascular junction, combined with aggressive vascular changes signaling increased retinal ischemia. According to ICROP classification, the presence of Plus in this posterior zone elevates the risk of neovascularization and vitreous hemorrhage, necessitating immediate intervention to prevent irreversible visual impairment. Decision: Treatment within less than 1 week using laser therapy to destroy avascular retina or anti-VEGF agents; coordinate with a retinal specialist for binocular indirect ophthalmoscopy follow-up, ensuring parental education on signs of progression like strabismus or leukocoria.
- **Zone: Zone 1, Plus: Plus, Stage: Stage 1, Decision: Treatment** Early Stage 1 ROP in Zone 1 with Plus disease shows a demarcation line separating vascularized from avascular retina, but the Plus component (arterial tortuosity and venous dilation) indicates severe hypoxia and imminent progression. ETROP studies highlight that Zone 1 Plus, even at low stages, correlates with poor outcomes without prompt action, including risks of macular involvement. Decision: Treatment within less than 1 week via cryotherapy or modern modalities like anti-VEGF; immediate specialist consultation is critical, with serial examinations to monitor for threshold disease and potential side effects such as endophthalmitis.
- **Zone: Zone 1, Plus: Plus, Stage: Stage 0, Decision: Treatment** Immature vascularization at Stage 0 in Zone 1 with Plus disease reflects incomplete retinal vessel development

accompanied by abnormal vascular caliber, suggesting high-risk pre-threshold ROP per ICROP. This configuration increases susceptibility to aggressive posterior ROP (AP-ROP), with potential for cicatricial changes leading to tractional detachment.

Decision: Treatment within less than 1 week, often starting with anti-VEGF to promote regression; refer urgently to neonatology-integrated ophthalmology services, emphasizing fluorescein angiography if needed for precise avascular zone mapping and long-term visual acuity surveillance.

- **Zone: Zone 1, Plus: Plus, Stage: Normal, Decision: Treatment** Normal staging in Zone 1 with Plus disease implies subtle vascular anomalies without overt ROP features, yet the Plus signs of tortuosity and dilation signal underlying ischemia in the posterior retina. Guidelines from AAP recommend early intervention in such cases to avert progression to fibrovascular membranes. Decision: Treatment within less than 1 week using targeted laser or pharmacologic approaches; specialist evaluation is imperative, including wide-field imaging to document baseline and track peripheral vascular maturation over subsequent weeks.
- **Zone: Zone 1, Plus: No Plus, Stage: Stage 3, Decision: Treatment** Stage 3 ROP in Zone 1 without Plus involves EFP extending into the vitreous, posing a substantial threat of macular detachment despite absent aggressive vascular signs. ETROP criteria classify this as Type 1 ROP due to the central location, with risks including retinal folds and vision loss. Decision: Treatment within less than 1 week via laser ablation of peripheral avascular areas; prompt referral to a vitreoretinal surgeon, with follow-up imaging to confirm EFP regression and monitor for secondary complications like glaucoma.
- **Zone: Zone 1, Plus: No Plus, Stage: Stage 2, Decision: Follow-up ≤ 1 week** Stage 2 ROP in Zone 1 without Plus presents as a ridge formation at the vascular terminus, indicating moderate ischemia in a high-risk posterior area per ICROP. While not yet threshold, close surveillance is needed to detect emerging Plus or proliferation, as delays can lead to tractional forces. Decision: Follow-up within ≤ 1 week using fundus examination; educate caregivers on prematurity-related risks, and prepare for potential escalation to treatment if changes occur, incorporating oxygen saturation monitoring in NICU settings.
- **Zone: Zone 1, Plus: No Plus, Stage: Stage 1, Decision: Follow-up ≤ 1 week** Early Stage 1 in Zone 1 without Plus shows a flat demarcation line, signifying initial vascular arrest

with potential for spontaneous resolution but heightened risk due to zonation. AAP screening protocols emphasize weekly checks to track progression metrics like clock hours involved. Decision: Follow-up within ≤ 1 week with detailed ophthalmoscopy; document postmenstrual age and birth weight for risk stratification, advising on environmental factors like supplemental oxygen that may influence ROP evolution.

- **Zone: Zone 1, Plus: No Plus, Stage: Stage 0, Decision: Follow-up 1–2 weeks** Stage 0 (immature, No ROP) in Zone 1 without Plus denotes incomplete vascularization without demarcation, common in very preterm infants but requiring vigilance for AP-ROP onset. Studies indicate that posterior immaturity correlates with higher incidence of severe ROP if unmonitored. Decision: Follow-up in 1–2 weeks to assess vessel extension; use digital retinal imaging for serial comparison, and integrate with multidisciplinary care including neonatologists for optimizing growth factors like IGF-1 levels.
- **Zone: Zone 1, Plus: No Plus, Stage: Normal, Decision: Follow-up 1–2 weeks** Normal mature vascularization in Zone 1 without Plus suggests stable retinal development, yet in premature contexts, subtle microvascular changes may persist. ICROP advises continued screening until full vascular maturity to rule out late-onset anomalies. Decision: Follow-up in 1–2 weeks for confirmatory examination; provide parental guidance on long-term visual assessments, potentially including electroretinography if family history of retinal disorders exists.
- **Zone: Zone 2, Plus: Plus, Stage: Stage 3, Decision: Treatment** Stage 3 ROP in Zone 2 with Plus features EFP with aggressive vascular signs, classifying as Type 1 ROP and risking peripheral detachment or plus-related complications like hemorrhage. ETROP trials support early treatment to improve structural outcomes. Decision: Treatment within less than 1 week using laser or anti-VEGF; urgent specialist involvement, with post-intervention fluorescein angiography to evaluate perfusion and prevent recurrence.
- **Zone: Zone 2, Plus: Plus, Stage: Stage 2, Decision: Treatment** Stage 2 in Zone 2 with Plus shows ridge elevation alongside vascular tortuosity, indicating threshold disease per guidelines and high progression potential to Stage 3 or beyond. This setup increases odds of cicatricial ROP leading to strabismus. Decision: Treatment within less than 1 week via ablative therapy; coordinate with pediatric retina experts, emphasizing anesthetic considerations in neonates and monitoring for systemic anti-VEGF effects.

- **Zone: Zone 2, Plus: Plus, Stage: Stage 1, Decision: Follow-up 1-2 week(s)** Stage 1 ROP in Zone 2 with Plus displays a demarcation line with mild aggressive features, warranting observation as it borders pre-threshold but may regress. ICROP notes variable outcomes, influenced by gestational age. Decision: Follow-up in 1-2 weeks for progression assessment; use wide-field cameras for documentation, and discuss nutritional interventions like omega-3 supplementation to support vascular health.
- **Zone: Zone 2, Plus: Plus, Stage: Stage 0, Decision: Follow-up 1-2 week(s)** Immature Stage 0 in Zone 2 with Plus suggests early vascular stress without structural ROP, but Plus elevates monitoring needs to catch rapid changes. Research links this to hyperoxia exposure in NICUs. Decision: Follow-up in 1-2 weeks with detailed funduscopy; integrate blood gas monitoring, and prepare educational materials for families on ROP risk factors.
- **Zone: Zone 2, Plus: Plus, Stage: Normal, Decision: Follow-up 1-2 week(s)** Normal vascular state in Zone 2 with Plus implies residual aggressive signs post-regression, requiring checks for subtle recurrences. AAP guidelines stress extended screening in such variants. Decision: Follow-up in 1-2 weeks to confirm stability; employ optical coherence tomography (OCT) if available for macular evaluation, and advise on follow-up vision therapy if needed.
- **Zone: Zone 2, Plus: No Plus, Stage: Stage 3, Decision: Follow-up <=1 week** Stage 3 in Zone 2 without Plus involves EFP but lacks aggressive vascularity, classified as Type 2 ROP with moderate detachment risk. ETROP recommends tight intervals to detect Plus onset. Decision: Follow-up within <=1 week using imaging; monitor clock-hour extent, and collaborate with endocrinologists for growth hormone influences on ROP.
- **Zone: Zone 2, Plus: No Plus, Stage: Stage 2, Decision: Follow-up 1-2 weeks** Stage 2 ROP in Zone 2 without Plus features ridge without proliferation, often regressing spontaneously but needing surveillance for threshold crossing. Studies show better prognosis with early detection. Decision: Follow-up in 1-2 weeks for vascular reassessment; document with photos, and consider anemia corrections as contributing factors.
- **Zone: Zone 2, Plus: No Plus, Stage: Stage 1, Decision: Follow-up 2 weeks** Stage 1 in Zone 2 without Plus shows simple demarcation, low-risk for progression in mid-zone per ICROP. Routine monitoring suffices for most cases. Decision: Follow-up in 2 weeks

to track line migration; educate on sepsis risks exacerbating ROP, and plan for discharge criteria once Zone 3 is reached.

- **Zone: Zone 2, Plus: No Plus, Stage: Stage 0, Decision: Follow-up 2-3 weeks** Immature Stage 0 in Zone 2 without Plus indicates ongoing vascularization without disease, suitable for longer intervals as per AAP. Focus on natural maturation. Decision: Follow-up in 2-3 weeks; use telemedicine if feasible for remote areas, and monitor weight gain as a proxy for retinal health.
- **Zone: Zone 2, Plus: No Plus, Stage: Normal, Decision: Follow-up 2-3 weeks** Normal maturity in Zone 2 without Plus suggests nearing screening endpoint, but confirm no peripheral issues. Guidelines allow extension based on PMA. Decision: Follow-up in 2-3 weeks for final check; discuss long-term amblyopia risks, and integrate with developmental pediatrics.
- **Zone: Zone 3, Plus: Plus, Stage: Stage 2, Decision: Follow-up 2-3 weeks** Stage 2 in Zone 3 with Plus features ridge in peripheral retina with mild aggression, rarely progressing but monitored for completeness. ICROP notes high regression rates here. Decision: Follow-up in 2-3 weeks; use fundus photography, and advise on light exposure controls in incubators.
- **Zone: Zone 3, Plus: Plus, Stage: Stage 1, Decision: Follow-up 2-3 weeks** Stage 1 ROP in Zone 3 with Plus shows peripheral demarcation with vascular signs, low urgency due to anterior location. Often self-resolves per studies. Decision: Follow-up in 2-3 weeks for confirmation; monitor for rare traction, and educate on genetic predispositions if relevant.
- **Zone: Zone 3, Plus: No Plus, Stage: Stage 2, Decision: Follow-up 2-3 weeks** Stage 2 in Zone 3 without Plus involves peripheral ridge, minimal risk of vision-threatening issues. ETROP supports extended follow-up. Decision: Follow-up in 2-3 weeks; prepare for screening termination if regressed, and note transfusion history as a modifier.
- **Zone: Zone 3, Plus: No Plus, Stage: Stage 1, Decision: Follow-up 2-3 weeks** Stage 1 in Zone 3 without Plus is the mildest form, with demarcation far from macula. High spontaneous resolution likelihood. Decision: Follow-up in 2-3 weeks; use simple ophthalmoscopy, and discuss vitamin E supplementation trials.
- **Zone: Zone 3, Plus: No Plus, Stage: Stage 0, Decision: Follow-up 2-3 weeks** Immature Stage 0 in Zone 3 without Plus nears full vascularization, signaling end of at-risk period.

AAP criteria for discharge apply. Decision: Follow-up in 2-3 weeks; confirm with imaging, and plan for routine pediatric eye exams post-discharge.

- **Zone: Zone 3, Plus: No Plus, Stage: Normal, Decision: Follow-up 2-3 weeks** Fully normal in Zone 3 without Plus indicates mature retina, ready for screening cessation. Ensure no missed peripherals. Decision: Follow-up in 2-3 weeks as final verification; provide certificates of completion, and advise on school-age vision screening.
- **Zone: Zone 1, Plus: No Plus, Stage: Stage 4, Decision: Treatment** Stage 4 ROP in Zone 1 without Plus involves partial retinal detachment, often tractional from prior proliferation, threatening central vision. Urgent surgical need per guidelines. Decision: Treatment within less than 1 week, potentially vitrectomy; mobilize surgical team, with pre-op OCT for extent mapping.
- **Zone: Zone 2, Plus: No Plus, Stage: Stage 4, Decision: Treatment** Stage 4 in Zone 2 without Plus features localized detachment, requiring intervention to prevent total involvement. Prognosis varies with macular status. Decision: Treatment within less than 1 week via scleral buckling or laser; specialist OR preparation, monitoring for amblyopia post-op.
- **Zone: Zone 3, Plus: No Plus, Stage: Stage 4, Decision: Treatment** Peripheral Stage 4 in Zone 3 without Plus shows anterior detachment, less vision-threatening but still needs addressal. Rare but actionable. Decision: Treatment within less than 1 week with targeted surgery; assess with ultrasound, and focus on rehabilitation potential.
- **Zone: Zone 1, Plus: No Plus, Stage: Stage 5, Decision: Treatment** Stage 5 ROP in Zone 1 without Plus is total funnel detachment, often with poor visual prognosis but salvage attempts warranted. Involves complex vitreoretinal surgery. Decision: Treatment within less than 1 week; urgent enucleation consideration if intractable, with psychological support for families.
- **Zone: Zone 2, Plus: No Plus, Stage: Stage 5, Decision: Treatment** Total Stage 5 detachment in Zone 2 without Plus signifies end-stage ROP, focusing on pain management and cosmesis. Surgical options limited. Decision: Treatment within less than 1 week via advanced procedures; involve low-vision specialists early for adaptive strategies.
- **Zone: Zone 3, Plus: No Plus, Stage: Stage 5, Decision: Treatment** Stage 5 in Zone 3 without Plus is peripheral total detachment, potentially sparing central vision if isolated.

Requires confirmation imaging. Decision: Treatment within less than 1 week with localized interventions; emphasize multidisciplinary follow-up for associated systemic prematurity issues.

Zone: Zone 1, Plus: Plus, Stage: Stage 4, Decision: Treatment Stage 4 ROP in Zone 1 with Plus disease indicates partial retinal detachment with aggressive vascular tortuosity and dilation, posing an imminent threat to central vision due to posterior involvement. The presence of Plus exacerbates ischemia and neovascularization, necessitating urgent surgical intervention beyond standard ablation. Decision: Treatment within less than 1 week, likely involving vitrectomy or scleral buckling; immediate referral to a vitreoretinal surgeon is critical, with pre-operative wide-field imaging to map detachment extent and post-op monitoring for reattachment success.

Zone: Zone 1, Plus: Plus, Stage: Stage 5, Decision: Treatment Stage 5 ROP in Zone 1 with Plus disease represents total retinal detachment with severe vascular abnormalities, often leading to a closed-funnel configuration and profound visual loss. The Plus component reflects advanced ischemia driving cicatricial changes, requiring aggressive salvage attempts despite poor prognosis. Decision: Treatment within less than 1 week via complex vitreoretinal surgery or enucleation if intractable; urgent specialist consultation is essential, with psychological support for families and planning for prosthetic fitting if needed.

Zone: Zone 2, Plus: Plus, Stage: Stage 4, Decision: Treatment Stage 4 ROP in Zone 2 with Plus disease features localized detachment accompanied by Plus-related vascular stress, increasing the risk of progression to total detachment in a mid-peripheral zone. The aggressive vascular signs suggest ongoing hypoxia, necessitating rapid surgical correction to preserve remaining vision. Decision: Treatment within less than 1 week, potentially with scleral buckling or vitrectomy; coordinate with a surgical team, using intraoperative OCT to guide repair and monitor for post-operative inflammation or recurrence.

Zone: Zone 2, Plus: Plus, Stage: Stage 5, Decision: Treatment Total Stage 5 detachment in Zone 2 with Plus disease indicates end-stage ROP with extensive cicatricial traction and vascular abnormalities, severely compromising retinal function. The Plus component reflects persistent ischemia, making surgical intervention a last resort to alleviate pain or improve cosmesis. Decision: Treatment within less than 1 week via advanced vitreoretinal procedures or enucleation; involve low-vision specialists early, with detailed pre-op ultrasound to assess detachment severity and family counseling on outcomes.

Zone: Zone 3, Plus: Plus, Stage: Stage 4, Decision: Treatment Stage 4 ROP in Zone 3 with Plus disease shows anterior partial detachment with abnormal vascular features, a rare but serious finding in the peripheral retina that requires intervention to prevent further progression. The Plus signs indicate ongoing neovascular drive, potentially affecting adjacent zones if untreated. Decision: Treatment within less than 1 week, likely with localized surgery or laser-assisted repair; urgent evaluation with B-scan ultrasound is needed, followed by rehabilitation planning for residual vision.

Zone: Zone 3, Plus: Plus, Stage: Stage 5, Decision: Treatment Stage 5 ROP in Zone 3 with Plus disease presents as total peripheral detachment with aggressive vascular changes, though less likely to impact central vision due to its anterior location. The Plus component suggests persistent hypoxia, necessitating intervention despite low visual salvage potential. Decision: Treatment within less than 1 week via targeted surgical approaches or pain management procedures; specialist assessment with imaging is urgent, with focus on multidisciplinary care for associated prematurity complications.